

LTMB12-SQL Practice - Set1

```
-- =====  
-- HOSPITAL MANAGEMENT DATABASE  
-- ALL TABLES + DATA IN ONE SCRIPT  
-- =====
```

```
CREATE DATABASE hospital_db;  
USE hospital_db;
```

```
-- =====  
-- PATIENTS  
-- =====
```

```
CREATE TABLE patients (  
    patient_id INT PRIMARY KEY,  
    patient_name VARCHAR(100),  
    gender VARCHAR(10),  
    age INT,  
    city VARCHAR(50)  
);
```

```
INSERT INTO patients VALUES  
(1,'Rahul','Male',35,'Chennai'),  
(2,'Priya','Female',28,'Bangalore'),  
(3,'Arun','Male',45,'Hyderabad'),  
(4,'Sneha','Female',31,'Mumbai'),  
(5,'Kiran','Male',50,'Delhi'),  
(6,'Ramesh Kumar','Male',42,'Chennai'),  
(7,'Lakshmi Devi','Female',36,'Coimbatore'),  
(8,'Suresh Babu','Male',29,'Madurai'),  
(9,'Anitha Rao','Female',55,'Bangalore'),  
(10,'Vijay Sharma','Male',61,'Mumbai'),  
(11,'Deepa Nair','Female',48,'Kochi'),  
(12,'Karthik Raj','Male',34,'Chennai'),  
(13,'Meena Patel','Female',40,'Ahmedabad'),  
(14,'Rohit Singh','Male',27,'Delhi'),  
(15,'Pooja Gupta','Female',31,'Hyderabad'),  
(16,'Naveen Kumar','Male',45,'Pune'),  
(17,'Shalini Verma','Female',52,'Lucknow'),  
(18,'Ajay Das','Male',38,'Kolkata'),  
(19,'Neha Reddy','Female',24,'Hyderabad'),  
(20,'Manoj Jain','Male',58,'Jaipur');
```

```
-- =====
```

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-- DOCTORS

-- =====

```
CREATE TABLE doctors (  
    doctor_id INT PRIMARY KEY,  
    doctor_name VARCHAR(100),  
    specialization VARCHAR(50),  
    consultation_fee DECIMAL(10,2)  
);
```

```
INSERT INTO doctors VALUES  
(101,'Dr. Kumar','Cardiology',1000),  
(102,'Dr. Mehta','Neurology',1500),  
(103,'Dr. Sharma','Orthopedics',1200),  
(104,'Dr. Reddy','Dermatology',800),  
(105,'Dr. Anjali Iyer','Pediatrics',900),  
(106,'Dr. Rajesh Gupta','ENT',850),  
(107,'Dr. Vivek Menon','General Medicine',700),  
(108,'Dr. Sunita Shah','Gynecology',1300),  
(109,'Dr. Akash Verma','Cardiology',1200),  
(110,'Dr. Harish Rao','Neurology',1800),  
(111,'Dr. Kavitha Devi','Orthopedics',1400),  
(112,'Dr. Prakash Nair','Pulmonology',1100);
```

-- =====

-- APPOINTMENTS

-- =====

```
CREATE TABLE appointments (  
    appointment_id INT PRIMARY KEY,  
    patient_id INT,  
    doctor_id INT,  
    appointment_date DATE,  
    status VARCHAR(20),  
  
    FOREIGN KEY(patient_id)  
        REFERENCES patients(patient_id),  
  
    FOREIGN KEY(doctor_id)  
        REFERENCES doctors(doctor_id)  
);
```

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```
INSERT INTO appointments VALUES
(1001,1,101,'2025-05-01','Completed'),
(1002,2,102,'2025-05-03','Completed'),
(1003,3,101,'2025-05-05','Pending'),
(1004,4,103,'2025-05-06','Completed'),
(1005,5,104,'2025-05-08','Cancelled'),
(1006,6,105,'2025-05-10','Completed'),
(1007,7,106,'2025-05-11','Completed'),
(1008,8,107,'2025-05-11','Completed'),
(1009,9,108,'2025-05-12','Pending'),
(1010,10,109,'2025-05-12','Completed'),
(1011,11,110,'2025-05-13','Completed'),
(1012,12,111,'2025-05-13','Completed'),
(1013,13,112,'2025-05-14','Cancelled'),
(1014,14,105,'2025-05-14','Completed'),
(1015,15,106,'2025-05-15','Completed'),
(1016,16,107,'2025-05-16','Pending'),
(1017,17,108,'2025-05-17','Completed'),
(1018,18,109,'2025-05-17','Completed'),
(1019,19,110,'2025-05-18','Completed'),
(1020,20,111,'2025-05-18','Pending'),
(1021,1,112,'2025-05-19','Completed'),
(1022,2,105,'2025-05-20','Completed'),
(1023,3,106,'2025-05-20','Completed'),
(1024,4,107,'2025-05-21','Completed'),
(1025,5,108,'2025-05-21','Cancelled');
```

```
-- =====
-- BILLING
-- =====
```

```
CREATE TABLE billing (
    bill_id INT PRIMARY KEY,
    appointment_id INT,
    bill_amount DECIMAL(10,2),
    payment_status VARCHAR(20),

    FOREIGN KEY(appointment_id)
        REFERENCES appointments(appointment_id)
);
```

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INSERT INTO billing VALUES

```
(1,1001,1000,'Paid'),
(2,1002,1500,'Paid'),
(3,1003,1000,'Pending'),
(4,1004,1200,'Paid'),
(5,1005,0,'Cancelled'),
(6,1006,900,'Paid'),
(7,1007,850,'Paid'),
(8,1008,700,'Paid'),
(9,1009,1300,'Pending'),
(10,1010,1200,'Paid'),
(11,1011,1800,'Paid'),
(12,1012,1400,'Paid'),
(13,1013,0,'Cancelled'),
(14,1014,900,'Paid'),
(15,1015,850,'Paid'),
(16,1016,700,'Pending'),
(17,1017,1300,'Paid'),
(18,1018,1200,'Paid'),
(19,1019,1800,'Paid'),
(20,1020,1400,'Pending'),
(21,1021,1100,'Paid'),
(22,1022,900,'Paid'),
(23,1023,850,'Paid'),
(24,1024,700,'Paid'),
(25,1025,0,'Cancelled');
```

```
-- =====
-- TREATMENTS
-- =====
```

```
CREATE TABLE treatments (
  treatment_id INT PRIMARY KEY,
  appointment_id INT,
  treatment_name VARCHAR(100),
  treatment_cost DECIMAL(10,2),

  FOREIGN KEY(appointment_id)
    REFERENCES appointments(appointment_id)
);
```

INSERT INTO treatments VALUES

```
(1,1001,'ECG',500),
(2,1002,'Brain Scan',2500),
```

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```
(3,1004,'Knee X-Ray',1200),
(4,1006,'Vaccination',300),
(5,1007,'Ear Cleaning',400),
(6,1008,'General Checkup',250),
(7,1010,'ECG',500),
(8,1011,'MRI Scan',3500),
(9,1012,'Bone Scan',1800),
(10,1014,'Vaccination',300),
(11,1015,'ENT Checkup',500),
(12,1017,'Pregnancy Scan',2200),
(13,1018,'Stress Test',900),
(14,1019,'Neurology Test',4000),
(15,1021,'Pulmonary Function Test',1600);
```

```
-- =====
-- MEDICINES
-- =====
```

```
CREATE TABLE medicines (
    medicine_id INT PRIMARY KEY,
    medicine_name VARCHAR(100),
    medicine_price DECIMAL(10,2)
);
```

```
INSERT INTO medicines VALUES
(1,'Paracetamol',20),
(2,'Amoxicillin',120),
(3,'Aspirin',50),
(4,'Metformin',180),
(5,'Atorvastatin',250),
(6,'Omeprazole',90),
(7,'Cetirizine',40),
(8,'Vitamin D',150);
```

```
-- =====
-- PRESCRIPTIONS
-- =====
```

```
CREATE TABLE prescriptions (
    prescription_id INT PRIMARY KEY,
    appointment_id INT,
    medicine_id INT,
    quantity INT,
```

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```
FOREIGN KEY(appointment_id)
REFERENCES appointments(appointment_id),
```

```
FOREIGN KEY(medicine_id)
REFERENCES medicines(medicine_id)
```

```
);
```

```
INSERT INTO prescriptions VALUES
```

```
(1,1001,1,10),
(2,1001,3,5),
(3,1002,2,7),
(4,1004,5,15),
(5,1006,7,10),
(6,1007,6,5),
(7,1008,1,10),
(8,1010,4,20),
(9,1011,5,30),
(10,1012,3,15),
(11,1014,8,10),
(12,1015,7,5),
(13,1017,2,10),
(14,1018,1,20),
(15,1019,5,15);
```

```
-- =====
```

```
-- APPOINTMENT AUDIT
```

```
-- =====
```

```
CREATE TABLE appointment_audit (
  audit_id INT AUTO_INCREMENT PRIMARY KEY,
  appointment_id INT,
  action_type VARCHAR(50),
  action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

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```
-- =====  
-- DATA SUMMARY  
-- =====  
-- Patients      : 20  
-- Doctors       : 12  
-- Appointments  : 25  
-- Billing        : 25  
-- Treatments    : 15  
-- Medicines     : 8  
-- Prescriptions : 15  
-- Audit Table   : Empty (for Trigger Practice)  
-- =====
```

This dataset is large enough to practice the following Topics:

- ☐ All Joins (INNER, LEFT, RIGHT, SELF)
- ☐ Aggregate Functions
- ☐ GROUP BY / HAVING
- ☐ Subqueries
- ☐ CTEs
- ☐ Window Functions (ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD)
- ☐ Views
- ☐ Stored Procedures
- ☐ Functions
- ☐ Triggers
- ☐ Case Statements
- ☐ Real-world Hospital Analytics and Reporting Queries.

Easy Level Questions (2 Marks Each × 5 = 10 Marks)

1. Inner Join

Scenario:

The hospital administrator wants to know which patients have successfully completed appointments and the doctors they consulted.

Question:

Write a query to display:

- Patient Name
- Doctor Name
- Appointment Date

for all appointments with status = 'Completed'.

2. Left Join

Scenario:

The finance team wants to identify all appointments and check whether a bill has been generated for each appointment.

Question:

Write a query to display:

- Appointment ID
- Appointment Date
- Bill Amount

Show all appointments even if billing information is missing.

3. Simple Subquery

Scenario:

Management wants to identify doctors who charge more than the average consultation fee.

Question:

Write a query to display:

- Doctor ID
- Doctor Name
- Consultation Fee

for doctors whose consultation fee is greater than the average consultation fee of all doctors.

4. Join with Aggregate Function

Scenario:

The hospital director wants to know how many appointments each doctor handled.

Question:

Write a query to display:

- Doctor Name
- Total Appointments

for every doctor.

5. Join with Aggregate Function + HAVING

Scenario:

The management wants to reward doctors who handled more than 2 appointments.

Question:

Write a query to display:

- Doctor Name
- Total Appointments

for doctors who handled more than 2 appointments.

Medium Level Questions (4 Marks Each × 5 = 20 Marks)

1. Join + Subquery

Scenario:

The medical director wants to identify patients who consulted doctors from the Cardiology department.

Question:

Write a query to display:

- Patient Name
- Doctor Name
- Appointment Date

Use a subquery to first identify all Cardiology doctors.

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2. Join + Aggregate + Subquery

Scenario:

Management wants to identify doctors whose total generated billing amount is greater than the average billing generated by all doctors.

Question:

Write a query to display:

- Doctor Name
- Total Revenue Generated

Only show doctors whose total revenue is above the average doctor revenue.

3. Join with Window Function (ROW_NUMBER)

Scenario:

The hospital wants to find the most recent appointment for every patient.

Question:

Write a query to display:

- Patient Name
- Appointment Date
- Doctor Name

Return only the latest appointment of each patient using ROW_NUMBER().

4. Join with Window Function (RANK)

Scenario:

The finance team wants to rank doctors based on total billing revenue generated.

Question:

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Write a query to display:

- Doctor Name
- Total Revenue
- Revenue Rank

Use RANK() to assign rankings based on revenue generated.

5. Complex Join + Aggregate + Window Function

Scenario:

The hospital CEO wants to identify the highest revenue-generating doctor within each specialization.

Question:

Write a query to display:

- Specialization
- Doctor Name
- Total Revenue

Return only the top revenue-generating doctor from each specialization using a window function.

SQL Joins + Window Functions (20 Scenario-Based Questions)

Dataset: Hospital Management System

These questions are written in a real-world business format suitable for assessments, interviews, and lab exercises.

1. Latest Appointment for Every Patient

Scenario

The hospital administration wants to send follow-up feedback forms to patients based on their most recent visit.

Question

Write a query to display the latest appointment details for every patient, including:

- Patient Name
- Appointment Date
- Doctor Name

Return only the most recent appointment for each patient.

2. Rank Doctors Based on Consultation Fees

Scenario

The HR department wants to analyze the fee structure of doctors across the hospital.

Question

Write a query to rank all doctors based on their consultation fees from highest to lowest.

Display:

- Doctor Name
 - Specialization
 - Consultation Fee
 - Rank
-

3. Highest Bill Generated by Each Patient

Scenario

The finance team wants to identify the largest bill paid by each patient.

Question

Write a query to display the highest bill amount generated by every patient.

Display:

- Patient Name
- Bill Amount
- Appointment Date

Return only the highest bill for each patient.

4. Rank Doctors by Total Revenue

Scenario

Hospital management wants to evaluate doctor performance based on revenue generated through appointments.

Question

Write a query to rank doctors based on the total billing revenue they have generated.

Display:

- Doctor Name
 - Total Revenue
 - Revenue Rank
-

5. Most Expensive Treatment Handled by Each Doctor

Scenario

The medical director wants to know the costliest treatment handled by every doctor.

Question

Write a query to display the most expensive treatment performed under each doctor.

Display:

- Doctor Name
- Treatment Name
- Treatment Cost

Return only one record per doctor.

6. Appointment Numbering for Patients

Scenario

The patient care team wants to track how many times each patient has visited the hospital.

Question

Write a query to assign a sequence number to every appointment of each patient based on appointment date.

Display:

- Patient Name
 - Appointment Date
 - Visit Number
-

7. Dense Ranking of Doctors by Revenue

Scenario

The CEO wants to create a leaderboard of doctors based on revenue generation.

Question

Write a query to assign dense rankings to doctors based on total billing revenue.

Display:

- Doctor Name
 - Revenue
 - Dense Rank
-

8. Previous Appointment Date of Every Patient

Scenario

The hospital wants to analyze the frequency of patient visits.

Question

Write a query to display each appointment along with the previous appointment date of the same patient.

Display:

- Patient Name
 - Current Appointment Date
 - Previous Appointment Date
-

9. Next Appointment Date of Every Patient

Scenario

The appointment scheduling team wants to predict upcoming patient visits.

Question

Write a query to display each appointment along with the next appointment date of the same patient.

Display:

- Patient Name
 - Current Appointment Date
 - Next Appointment Date
-

10. Highest Revenue Doctor in Each Specialization

Scenario

The hospital board wants to recognize the top-performing doctor from every specialization.

Question

Write a query to identify the doctor who generated the highest billing revenue within each specialization.

Display:

- Specialization
 - Doctor Name
 - Total Revenue
-

11. Rank Patients Based on Total Hospital Spending

Scenario

The finance department wants to identify high-value patients based on overall spending.

Question

Write a query to rank patients according to the total amount billed across all their appointments.

Display:

- Patient Name
 - Total Billing Amount
 - Rank
-

12. Latest Treatment Received by Each Patient

Scenario

The medical records department wants to maintain the latest treatment history of every patient.

Question

Write a query to display the most recent treatment received by each patient.

Display:

- Patient Name
 - Treatment Name
 - Appointment Date
-

13. Top 2 Revenue-Generating Doctors

Scenario

The hospital management wants to award the best-performing doctors of the month.

Question

Write a query to identify the top 2 doctors based on total revenue generated.

Display:

- Doctor Name
 - Total Revenue
 - Rank
-

14. Running Revenue Generated by Each Doctor

Scenario

The finance team wants to track how revenue accumulates over time for every doctor.

Question

Write a query to display the cumulative billing revenue generated by each doctor after every appointment.

Display:

- Doctor Name
 - Appointment Date
 - Bill Amount
 - Running Revenue
-

15. Difference Between Current and Previous Bill

Scenario

The hospital wants to study spending patterns of repeat patients.

Question

Write a query to display the difference between the current bill amount and the previous bill amount for every patient.

Display:

- Patient Name
- Appointment Date
- Current Bill
- Previous Bill
- Difference

16. Most Frequently Visited Doctor for Each Patient

Scenario

The hospital wants to understand patient loyalty towards doctors.

Question

Write a query to identify the doctor whom each patient has visited the most.

Display:

- Patient Name
- Doctor Name
- Total Visits

Return only one doctor per patient.

17. Rank Treatments by Cost Within Each Doctor

Scenario

The operations team wants to analyze the treatment portfolio of each doctor.

Question

Write a query to rank treatments handled by each doctor based on treatment cost.

Display:

- Doctor Name
- Treatment Name
- Treatment Cost
- Treatment Rank

18. First Appointment of Every Patient

Scenario

The hospital wants to study patient onboarding trends.

Question

Write a query to display the first appointment recorded for every patient.

Display:

- Patient Name
- Appointment Date
- Doctor Name

Return only the earliest appointment for each patient.

19. Doctor Revenue Contribution Percentage

Scenario

Management wants to understand how much each doctor contributes to the hospital's overall revenue.

Question

Write a query to calculate the percentage contribution of each doctor to the total hospital billing revenue.

Display:

- Doctor Name
- Revenue Generated
- Revenue Contribution %

20. Top 3 Patients by Spending in Each City

Scenario

The hospital marketing team wants to identify the most valuable patients in every city.

Question

Write a query to find the top 3 patients based on total billing amount within each city.

Display:

- City
- Patient Name
- Total Billing Amount
- Rank within City

Return only the top 3 patients from every city.
